Curriculum Vitae

Matthieu Pélissier

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Cosmologie Observationnelle, LPSC
Grenoble, France

Research Interests

Experimental and observational cosmology; probing dark matter with stellar streams; analysis of photometric survey data.

Education

2024–present	Ph.D. in Physics, LPSC Supervised by Dr. Marina Kuna and Dr. David Maurin, *Probe dark matter substructures with stellar streams detected by Rubin observatory.*
2022–2024	Master's in Particle Physics and Cosmology, University of Grenoble Alpes
2022–2024	Magister's in Fundamental Physics, University of Grenoble Alpes
2019–2022	B.Sc. in Physics, University of Toulouse III

Research Experience

2025	Collaboration Visit, 2 months, University of Guildford, UK Worked with Dr. Denis Erkal on stellar stream simulations.
2024	Internship, 5 months, LPSC Cosmology Team, France Supervised by Dr. Marine Kuna, *Probed dark matter using stellar streams.*
2023	Internship, 2 months, LPSC ATLAS Team, France Supervised by Dr. Pierre-Antoine Delsart, *Solved a differential equation using a neural network for ATLAS calibration.*
2022	Internship, 3 months, University of Turku, Finland Supervised by Dr. Alexandra Veledina, *Studied polarization of light from a supermassive black hole accretion disk.*

Honors and Awards

2025	IDEX International Mobility grant (UGA)
2024	PhD Funding (IN2P3/CNRS) Awarded via exam at the Grenoble Doctoral School of Physics.
2024	Excellence Grant IN2P3 (CNRS) Inaugural Reynald Pain “Deux Infinis” grant, including a funded internship.

Scientific Collaborations

2024–present	DESC Member of stellar stream topical team within the Dark Matter working group.

Selected Publications

2026	***M. Pélissier**, P. S. Ferguson, A. Drlica-Wagner, M. Kuna, D. Maurin*, “Impact of LSST systematics on stellar-stream density fluctuations for dark matter”, *eprint 2609.10897*
2025	*A. Veledina,* ***M. Pélissier***, “Analytical ray-tracing of synchrotron emission around accreting black holes”, *Astronomy & Astrophysics*, vol. 693.

Selected Talks

Sep. 2026	*Invited Talk:* Near Field Cosmology group meeting, Univ. Washington
Sep. 2026	Dark matter probe combination workshop, KICP, Chicago
Aug. 2026	Collaboration-Wide presentation, DESC
Jul. 2026	DESC meetting Dark Matter session, Paris/Boston
Mai. 2026	LSST France - LPSC, Annecy
April. 2026	LPSC PhD seminar, LPSC, Grenoble
Nov. 2025	*Invited Talk:* DESC stellar stream Workshop, Univ. Washington *StreamSim tutorial*

Nov. 2025	LSST France - LPSC, Grenoble
Jun. 2025	LSST France - IJClab, Saclay
Nov. 2024	LSST France - APC, Paris
Jun. 2024	LSST France - CPPM, Marseille

Posters

Aug. 2026	*Summer school:* Rodolphe Clédassou, Pornichet, France
Mar. 2026	*Conference:* Rencontres de Moriond, La Thuile, Italy
Jul. 2024	Journée du laboratoire, LPSC, Grenoble

Public Outreach

2026	Cosmology & PhD Careers Talk, Highschool Students
2026	Pint of science festival, general public
2025	Cosmology & PhD Careers Talk, Middle-School Students
2025	Public "First Light of LSST" Talk, Elementary Students Presented LSST first light results to children.
2024	Cosmology & PhD Careers Talk, Middle-School Students

Teaching Experience

2026	Teaching Assistant, Special Relativity (2nd-Year Bachelor) Conducted exercises sessions.
2025–2026	Teaching Assistant, Lagrangian and Hamiltonian Mechanics (3rd-Year Bachelor) Conducted exercises sessions, completed course resources

2025	Teaching Assistant, Muons Practical Work (3rd-Year Bachelor) Supervised data acquisition and analysis of muon velocity measurements.
2024–2025	Teaching Assistant, Python (3rd-Year Bachelor) Explained Python fundamentals, completed course resources and supervised practical exercises and student projects.

Responsabilities

2026–present	Founder, Post-PhD Career Portrait Series Interviews with former PhD students on career paths beyond academia, compiled as a resource for current doctoral students.
2026–present	Co-author of the guidance document for non-permanent members of the laboratory
2026–present	Coordinator group science meeting Weekly discussion of laboratory cosmology group
2026–present	Co-founder and Coordinator, Astrophysics Discussion Group Weekly sessions for presenting research, discussing papers, and sharing useful research tools for PhD students and newcomers.
2026	LOC member of the LSST France Weekly sessions for presenting research, discussing papers, and sharing useful research tools for PhD students and newcomers.
2025-2026	Co-Organizer, Laboratory PhD Seminar Series Research talks for graduate students.

Additional Training

2026	DESC Dark Energy School Vera C. Rubin Observatory and cosmology
2026	School of Statistics Adanced statistical and machine learning methods for subatomic physics, astroparticle physics, and cosmology.
2025-2026	Euclid Summer School Four weeks training about cosmological challenges with current and future surveys
2024-2026	Doctoral school training • How to teach training • Role and positioning of research in the face of environmental crises • Ethics of science training